

Social Media Engagement Analysis

Project Objective

- To evaluate **social media engagement patterns** across different content types and categories.
- To analyze the relationship between **content attributes and engagement metrics**.
- To identify which post types and strategies generate the highest engagement.
- To compare performance differences based on:
 - Content type
 - Country
 - Sentiment categories
 - User demographics
- To derive insights for **optimizing social media strategy and improving audience engagement**.

2. Data Sources

Source Description	Details
Dataset Name	Social Media Engagement Dataset
Sample Size	5,000 records
Domain	Social Media Analytics / Digital Marketing
Source	Synthetic dataset generated for analysis (GitHub repository)

3. Problem Statement

To analyse **post-type wise distribution** of engagement metrics (likes, comments, shares, engagement rate).

- To compare **content category performance** across different post categories (Food, Music, Travel, Fitness, Fashion).
- To evaluate the **impact of demographic factors** (age, gender, country) on engagement rates.
- To analyse the **performance of verified vs non-verified accounts** based on engagement scores.
- To study **device-wise contribution** (Mobile, Desktop, Tablet) to watch time and impressions.
- To study the **trend of engagement over time** using date-wise posted_at data.
- To analyse the **relationship between sentiment** (Positive, Negative, Neutral) and engagement metrics.
- **Seasonal and daily patterns** will be identified by extracting day names from the date field to understand variations in audience activity.

4. Attribute (Column/Features) Details

Fact Table / Main Dataset Details

Column Name	Data Type	Description
post_id	Integer	Unique identifier for each social media post
posted_at	DateTime	Date and time when the post was published
age	Integer	Age of the user/viewer (in years)
gender	Text (String)	Gender of the user (Male, Female, Other)
country	Text (String)	Country of the user (India, USA, Germany, Canada, Japan, France)
post_type	Text (String)	Type of content posted (Image, Video, Reel, Text)

Column Name	Data Type	Description
post_category	Text (String)	Category of the post (Food, Music, Travel, Fitness, Fashion)
device_type	Text (String)	Device used to view the content (Mobile, Desktop, Tablet)
likes	Integer	Number of likes received on the post
comments	Integer	Number of comments received on the post
shares	Integer	Number of shares received on the post
watch_time_sec	Integer	Total watch time in seconds (for video/reel content)
impression_count	Integer	Total number of impressions (views) of the post
follower_count	Integer	Follower count of the content creator
engagement_rate	Decimal	Percentage engagement rate (calculated from interactions)
hashtags	Text (String)	Hashtags used in the post
sentiment	Text (String)	Sentiment of the post (Positive, Negative, Neutral)
is_verified	Boolean	Whether the account is verified (True/False)

Derived Fields (Created During Analysis)

Column Name	Data Type	Description
Engagement_Score	Integer	Sum of likes + comments + shares (total interactions)
hashtag_count	Integer	Number of hashtags used in the post
day_name	Text (String)	Name of the day (Monday, Tuesday, etc.) extracted from posted_at

5. Tools & Technologies

Tool/Technology	Purpose
Python (Pandas)	Data cleaning, transformation, and manipulation
Python (NumPy)	Numerical operations and calculations
Matplotlib	Static data visualizations (scatter, line, bar, pie, histogram, box plots)
Seaborn	Statistical visualizations (count plots, bar plots, violin plots, pair plots, heatmaps)
Plotly Express	Interactive visualizations (line charts, bar charts, scatter plots)
Google Colab	Development environment (Jupyter notebook-based)

6. Data Pre-Processing in Python (Colab)

Original dataset consisted of 5,000 records which was processed for data cleaning and transformation.

python

```
import pandas as pd

df =
pd.read_csv('https://github.com/GeethaGunasekaran1/Dataset_rep/raw/refs/heads/main/social
_media_engagement_5000.csv')

df['posted_at'] = pd.to_datetime(df['posted_at'])
```

- CSV file imported using Pandas read_csv() function
- posted_at column converted to DateTime data type for time-based analysis
- Data types verified using df.info()

Missing Value Handling:

Column	Method Applied
age	Filled with median value, converted to integer
gender	Filled with mode (most frequent value)
likes	Filled with median value, converted to integer
comments	Filled with median value, converted to integer
shares	Filled with median value, converted to integer
sentiment	Filled with mode (most frequent value)

Data Formatting:

- post_type and post_category standardized using .str.capitalize()
- device_type and sentiment capitalized using lambda function
- All categorical columns converted to consistent string format

Duplicate Handling:

- Checked for duplicates in post_id column
- Removed duplicate entries using drop_duplicates(subset='post_id')

- Verified no remaining duplicates

Outlier Treatment (Engagement Rate):

- Calculated Q1 (25th percentile) and Q3 (75th percentile)
- Computed $IQR = Q3 - Q1$
- Defined lower bound = $Q1 - 1.5 \times IQR$
- Defined upper bound = $Q3 + 1.5 \times IQR$
- Removed outliers outside these bounds
- Skewness improved after outlier removal

Feature Engineering:

- **Hashtag Count:** Created by splitting hashtags string and counting words
 - **Engagement Score:** Created as sum of likes + comments + shares
 - **Day Name:** Extracted from posted_at using .dt.day_name()
-

7. Data Exploration & Analysis (Pandas)

Dataset Structure Overview:

Method	Output
head()	First 10 rows of dataset
tail()	Last 10 rows of dataset
shape	(Number of rows, Number of columns)
columns	List of all column names
info()	Data types and non-null counts
describe()	Statistical summary of numeric columns

Categorical Distribution Analysis:

Column	Unique Values
gender	Male, Female, Other
country	6 countries (India, USA, Germany, Canada, Japan, France)
post_type	Image, Video, Reel, Text
post_category	Food, Music, Travel, Fitness, Fashion
device_type	Mobile, Desktop, Tablet
sentiment	Positive, Negative, Neutral

Correlation Matrix:

- Created correlation matrix for numeric fields
- Identified relationships between likes, comments, shares, impressions, followers, and engagement rate
- Rounded to 2 decimal places for readability

GroupBy Summaries Performed:

- Average likes by post type
- Average impressions by country
- Average engagement rate by device type
- Average comments by post category

8. Data Wrangling

New Fields Created:

Field Name	Formula
Engagement_Score	likes + comments + shares
Hashtag_Count	Length of split hashtags string

DataFrame Operations:

- Used `pd.concat()` to combine original dataframe with new derived fields
- Verified column names after concatenation

Advanced GroupBy Summaries:

Grouped By	Aggregated Metric
post_type	Mean Engagement_Score (sorted descending)
country	Mean Engagement_Score (sorted descending)
sentiment	Mean Engagement_Score (sorted descending)
post_category	Sum of Hashtag_Counts (sorted descending)

Percentiles Calculated (25%, 50%, 75%, 90%):

- Applied to all engagement metrics
- Helped understand data distribution and identify performance benchmarks

Skewness & Kurtosis (Optional):

- Calculated for likes, comments, shares, `watch_time_sec`, `impression_count`
- Skewness values indicated direction of data distribution asymmetry

11. Key Insights & Findings

Content Performance Insights

Post Type Engagement Analysis:

Post Type	Average Engagement Rate
Image	29.4% (Highest)
Text	Moderate
Video	Moderate
Reel	Moderate

Insight: Image posts achieved the highest average engagement rate (29.4%), indicating strong audience interest in visual content. Text posts also performed well, showing users engage with informative or relatable written content.

Best-Performing Content Categories:

Rank	Category	Engagement Score
1	Food	Highest
2	Music	Very High
3	Travel	Very High
4	Fitness	Moderate
5	Fashion	Moderate

Insight: Food posts achieved the highest engagement score, indicating strong audience interest in food-related content. Music and Travel categories also performed very well, showing users actively engage with entertaining and experience-based posts.

Country-wise Engagement Rate:

Rank	Country	Engagement Rate
1	Germany	Highest
2	Canada	Strong
3	Japan	Strong
4	India	Moderate
5	USA	Moderate
6	France	Lowest

Insight: Germany recorded the highest average engagement rate, indicating users from this region interact more actively with content. Canada and Japan also showed strong engagement performance compared to other countries.

User Trends Insights

Age vs Engagement Score:

Age Group	Engagement Level
Age 49	Highest
Teenagers	Strong
30s	Strong
Middle-aged groups	Lower

Insight: Engagement levels varied across age groups, with users aged 49 showing the highest engagement score. Teenagers and users in their 30s also demonstrated strong engagement compared to other age groups.

Verified vs Non-Verified Accounts:

Account Type	Average Engagement Score
Non-Verified	Higher
Verified	Lower

Insight: Non-verified accounts achieved a higher average engagement score compared to verified accounts. This suggests account verification alone does not guarantee higher engagement performance; audiences may interact more with regular or relatable creators.

Behavioral Insights

Best Day for Impressions:

Rank	Day	Average Impressions
1	Tuesday	Highest
2	Saturday	Strong
3	Sunday	Strong
4	Thursday	Lowest

Insight: Tuesday recorded the highest average impression count, indicating users were most active on this day. Saturday and Sunday also showed strong impression performance, suggesting higher audience engagement during weekends.

Device Type Impact on Watch Time:

Device Type	Average Watch Time
Mobile	Highest
Tablet	Moderate

Device Type	Average Watch Time
Desktop	Lowest

Insight: Mobile users recorded the highest average watch time, indicating audiences prefer consuming content on mobile devices. Desktop devices had the lowest average watch time, suggesting lower viewing engagement compared to portable devices.

Sentiment Analysis Insights

Sentiment vs Engagement Score:

Sentiment	Engagement Score
Positive	Highest
Negative	Strong
Neutral	Lowest

Insight: Positive sentiment posts achieved the highest engagement score, indicating audiences respond better to uplifting or favorable content. Negative sentiment posts also generated strong engagement, possibly due to emotional reactions and discussions.

Negative vs Neutral Post Behavior:

Sentiment	Likes	Comments	Shares	Engagement Score
Negative	Higher	Moderate	Higher	Higher
Neutral	Lower	Slightly Higher	Lower	Lower

Insight: Negative sentiment posts achieved higher overall engagement scores than neutral posts. Negative posts received more likes and shares, suggesting emotionally intense content attracts greater attention.

12. Conclusions

- 1 **Image and Reel-related posts** generated the highest audience engagement across the platform
- 2 **Germany, Canada, and Japan** showed stronger engagement rates compared to other countries
- 3 **Mobile users** contributed the highest average watch time, indicating strong mobile content consumption
- 4 **Tuesdays and weekends** recorded the highest impression counts, reflecting peak audience activity periods
- 5 **Positive and negative sentiment posts** outperformed neutral content, showing emotionally engaging posts attract more user interaction
- 6 **Non-verified accounts** achieved higher engagement than verified accounts, suggesting authenticity matters more than verification status
- 7 **Food, Music, and Travel categories** consistently performed well across multiple engagement metrics
- 8 **Age 49 and teenage groups** showed the highest engagement scores, indicating key target demographics

13. Recommendations

Based on the analysis findings, the following recommendations are proposed:

Recommendation	Rationale
Prioritize Image and Reel content	These formats generate the highest engagement rates (29.4% for images)
Schedule posts on Tuesdays and weekends	Peak impression days for maximum content visibility
Optimize content for mobile devices	Mobile users have the highest watch time
Create emotionally engaging content	Positive and negative sentiments drive higher interaction than neutral content
Target Germany, Canada, and Japan	These countries show highest engagement rates
Focus on Food, Music, and Travel categories	Consistently top-performing content categories
Don't rely solely on verification status	Non-verified accounts can achieve higher engagement with quality content
Target age groups 49 and teenagers	Highest engagement scores from these demographics

14. Output Files

File Name	Description
social_media_cleaned_dataset.csv	Final cleaned dataset after all preprocessing, outlier treatment, and feature engineering